FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Main)-2019

Time Allotted: 3 Hours **Maximum Marks: 360**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  Section-A (01 – 30, 31 – 60, 61 – 90) *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

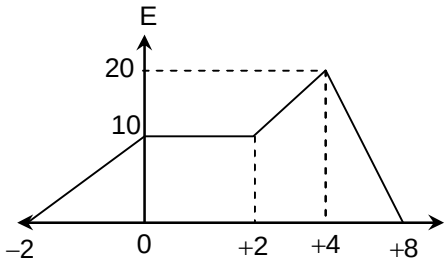
Physics

PART – I

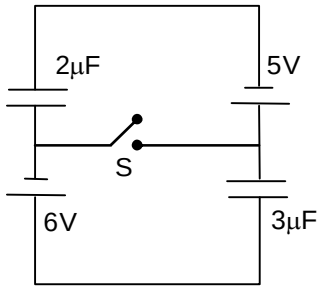
SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. If Electric field directed parallel to axis is represented by graph, and potential at origin is taken as zero. Find potential at x = 8 m
(A) 90 V
(B) 100 V
(C) 80 V
(D) None

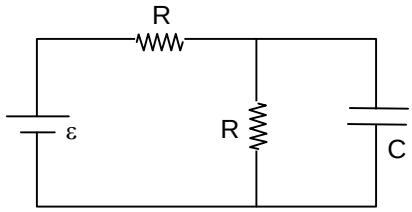


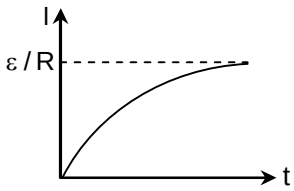
2. In the circuit shown switch S is kept open for long time. Switch is closed at t = 0, find total heat produced after closing the switch.
(A) $40\mu J$
(B) 89μJ
(C) 302.5μJ
(D) None

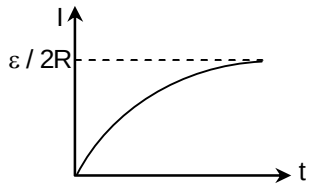


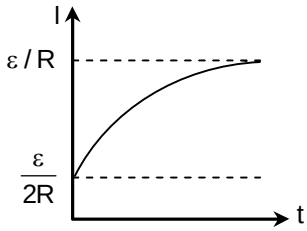
Space for Rough work

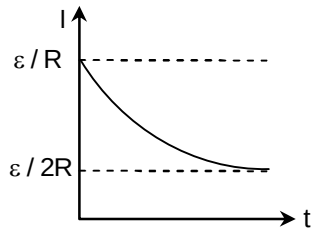
3. Consider the circuit shown in figure. If the switch is closed at t = 0. Then graph of current provided by battery as function of time is best represented by



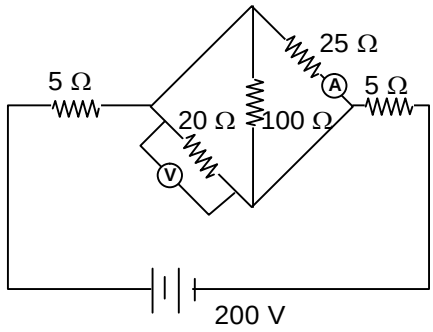
(A) 

(B) 

(C) 

(D) 

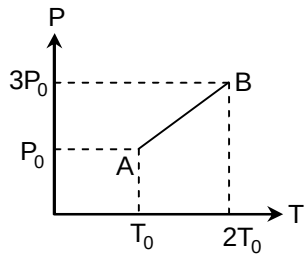
4. In the figure shown voltmeter and Ammeter are ideal. Reading of voltmeter and Ammeter respectively are
(A) 125 V, 3 A
(B) 120 V, 4 A
(C) 100 V, 4 A
(D) None



5. Pressure versus temperature graph of an ideal gas is shown in figure. If density of gas at point A is ρ_0, what is density at point B?

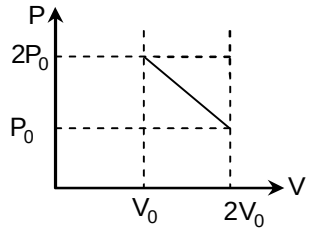
(A) $\frac{3}{4}\rho_0$ (B) $\frac{3}{2}\rho_0$

(C) $\frac{4}{3}\rho_0$ (D) $2\rho_0$



Space for Rough work

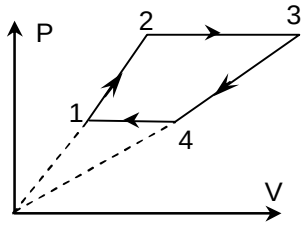
6. n moles of an ideal gas undergoes a thermodynamic process, in which pressure of gas varies linearly with volume as shown in figure. Find the maximum temperature of the gas during the process



(A) $\frac{P_0V_0}{nR}$ (B) $\frac{4P_0V_0}{nR}$

(C) $\frac{2P_0V_0}{nR}$ (D) $\frac{9P_0V_0}{4nR}$

7. Three moles of an ideal monoatomic gas perform a cyclic process as shown in figure. The gas temperature in different states are T_1 = 400 K, T_2 = 800 K, T_3 = 2400 K and T_4 = 1200 K. Find the work done during the cycle



(A) 12000 R (B) 2400 R

(C) 1200 R (D) 7200 R

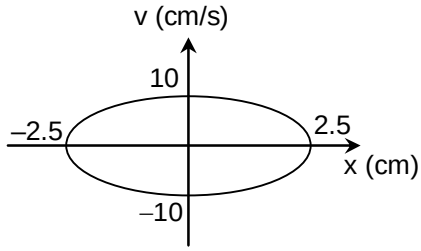
8. Time period of small oscillations of the two blocks shown in figure is



(A) $2\pi\sqrt{\frac{3m}{8k}}$ (B) $2\pi\sqrt{\frac{3m}{4k}}$

(C) $2\pi\sqrt{\frac{3m}{2k}}$ (D) None

9. If graph between velocity and displacement of a particle performing SHM is shown. Find the acceleration when the particle is at half of maximum displacement from mean position.



(A) 10 cm/s^2

(B) 20 cm/s^2

(C) 30 cm/s^2

(D) 40 cm/s^2

Space for Rough work

10. A wire of length 98 cm is stretched between two ends and extends by 2 cm. A organ pipe 50 cm long and open at both ends, when sounded with this stressed wire produces 8 beats in its fundamental mode. By decreasing stress beat frequency decreases. If density of wire is 10^4 Kg/m^3 and speed of sound in air is 292 m/s find young's modulus.

(A) $8.8\times10^{10}\,N/m^2$ (B) $4.4\times10^{10}\,N/m^2$

(C) $17.64\times10^{10}\,N/m^2$ (D) None

11. In order to detect the speed of an approaching aeroplane microwaves of wavelength 0.1 m are sent towards approaching plane and it is found that reflected waves produce 990 beats per second. The speed of the aeroplane is close to

(A) 20 m/s (B) 50 m/s

(C) 35 m/s (D) None

12. From the point on the ground at a distance 30 m from the foot of a pole a ball is thrown at an angle of 45°, which just touches the top of the pole and lands further 60 m from the foot of the pole on other side, height of the pole is

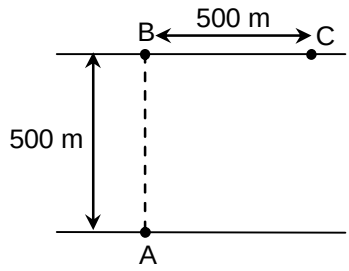
(A) 30 m (B) 45 m

(C) 22.5 m (D) 20 m

13. A river is flowing with a speed of 1 m/s. A swimmer wants to go to point C directly from point A. If he swims with a speed of 5 m/s at an angle θ with respect to water flow find θ.

(A) 37° (B) 60°

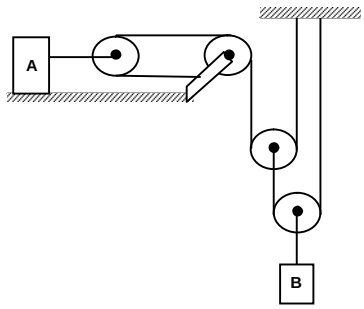
(C) 53° (D) 45°



14. In the figure shown if mass of each block is m. Find the acceleration of block A.

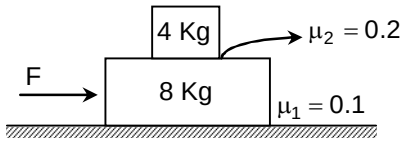
(A) 2 m/s^2 (B) 4 m/s^2

(C) 1 m/s^2 (D) None



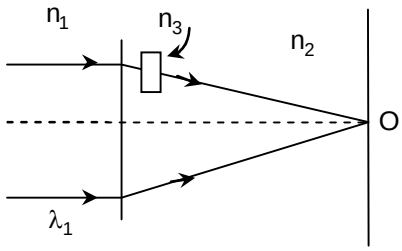
Space for Rough work

15. What is the minimum force at which any part of the system starts sliding



(A) 6 N (B) 9 N
(C) 12 N (D) None

16. In YDSE a parallel beam of light is incident on the slit from a medium of refractive index n_1. A transparent slab having thickness t is introduced in the path of one of the interfering beam. Retractive index of slab is n_3 and medium between slits and screen is of refractive index n_2. If wavelength of light in medium of refractive index n_1 is λ_1. Find phase difference at 'O'.



(A) $\frac{2\pi}{n_2\lambda}(n_3 - n_2)t$

(B) $\frac{2\pi}{n_1\lambda}(n_3 - n_1)t$

(C) $\frac{2\pi}{n_1\lambda}(n_3 - n_2)t$

(D) None

17. An unpolarized beam of intensity I_o is incident on a pair of polaroid making an angle 60° with each other. The intensity of light emerging from the pair is

(A) $I_o / 2$ (B) $I_o / 4$

(C) $I_o / 8$ (D) $\frac{3I_o}{4}$

18. The wavelength of K_α line from an element of atomic number 51 is λ. From another element the wavelength of K_α line is 4λ. What is the atomic number of second element?

(A) 25 (B) 26
(C) 100 (D) 99

Space for Rough work

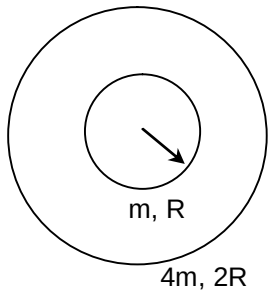
19. On illuminating the surface of a certain metal by two light of wavelengths $\lambda_1 = 350$ nm and $\lambda_2 = 540$ nm alternately, it is found that the ratio of maximum speeds in two cases comes out to be 2. Work function of metal is
(A) 3.22 eV (B) 5.65 eV
(C) 1.88 eV (D) None

20. Which of the following are known as universal gates?
(A) AND & OR (B) NAND & NOR
(C) NAND only (D) None

21. When the modulation percentage is 75% an AM transmitter radiates 10 KW power. How much of these is carries power?
(A) 7.8 KW (B) 7.5 KW
(C) 5 KW (D) None

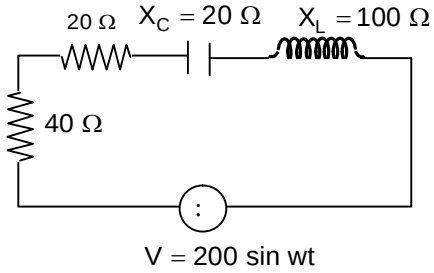
22. A rod of mass 3 kg and length 1 m is lying along the Y axis such that one of its end is at the origin. An Impulse is given to the rod due to which its end at origin acquire a velocity of 4 i and other end acquires a velocity 8 i. What is magnitude of angular momentum of rod about origin.
(A) 8 kmg^2/s (B) 18 kgm^2/s
(C) 10 kgm^2/s (D) None

23. What is gravitational potential energy of system consisting of two concentric spherical shells of mass, m, and 4 m and radii R and 2 R.
(A) $\frac{-2Gm^2}{R}$ (B) $\frac{-9Gm^2}{2R}$
(C) $-\frac{5Gm^2}{2R}$ (D) $-\frac{13Gm^2}{2R}$

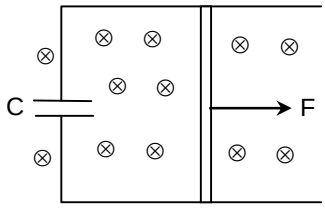


Space for Rough work

24. In figure shown the expression for potential difference across inductor is given by
(A) 200 sin wt
(B) 200 cos wt
(C) 200 sin (wt + 37°)
(D) 200 sin (wt + 53°)

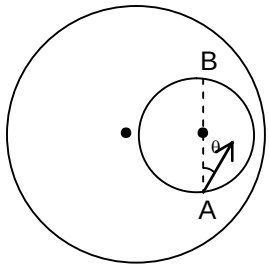


25. A uniform wire of mass m and length l slides freely along parallel rails on application of force F which starts acting at t = 0. Resistance of circuit is zero system is placed in uniform magnetic field B, perpendicular to plane of figure. What is current through capacitor at any time t. (Ignore any inductance)



(A) $\frac{BlCF}{m + B^2 l^2 C}$
(B) $\frac{BlCF}{m}t$
(C) $\frac{BlCFt^2}{m}$
(D) $\frac{BlCFt^2}{m + B^2 l^2 C}$

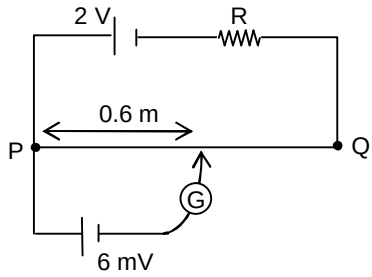
26. A long thick solid conductor has a cylindrical cavity of radius 1 m parallel to its axis at a distance of 2 m cylinder with cavity has a surface current density $\vec{J} = \frac{10^6}{\pi} A/m^2\ \hat{J}$. A charged particle of mass 0.1 kg and charge π coulomb is projected with velocity v at an angle of 60° as shown in figure. For what maximum v, particle is able to strike point B?



(A) 8 m/s
(B) 6 m/s
(C) 4 m/s
(D) None

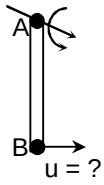
Space for Rough work

27. In the shown potentiometer arrangement the wire PQ has a resistance of $5\ \Omega$ and the driver cell has an emf of 2V. If a balance point is obtained at 0.6 m along PQ, what is the value of resistance R? (Length PQ = 1 m)



(A) 495 (B) 995 Ω
(C) 1495 Ω (D) 1995 Ω

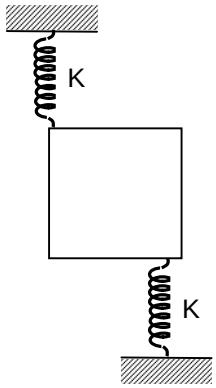
28. A particle of mass m is attached at the end B of uniform rod of same mass m and length ℓ. If the rod is whirled in vertical circle about end A, what is minimum speed of particle at end B, required to move it in vertical circle?



(A) $\sqrt{4gL}$ (B) $\sqrt{5gL}$
(C) $\sqrt{6gL}$ (D) None

29. A physical quantity a depends on two other physical quantities X and Y as $A = X^4Y^{1/4}$. If percentage error in the measurement of A and Y are 2% and 4% respectively. Then percentage error in the measurement of X is
(A) less than 1% (B) more than 1% but less than 2%
(C) between 2% and 4% (D) more than 4%

30. A square plate of mass m and side length is a kept as shown in figure. In equilibrium two of its side are horizontal and too vertical. What is its angular acceleration just after lower spring is cut.



(A) g/a (B) 2g/a
(C) 3g/2a (D) 4g/a

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The radial probability distribution curve of an orbital of H has $\dot{\dot{4}}$ local maxima. If orbital has $\dot{\dot{3}}$ angular node then orbital will be
(A) 7 f (B) 8 f
(C) 7 d (D) 8 d

32. NX is produced by the following step of reaction
$M + X_2 \rightarrow MX_2$
$3MX_2 + X_2 \rightarrow M_3X_8$
$M_3X_8 + N_2CO_3 \rightarrow NX + CO_2 + M_3O_4$
How much M(Metal) is consumed to produce 206 g of NX? (take at. wt. of M = 56, N = 23; X= 80)
(A) 42 gm (B) 56 gm
(C) $\frac{14}{3}$gm (D) $\frac{7}{4}$gm

33. On heating vapours of S_8 (g) decomposes to S_2 (g). Due to this, the Vander Waal's constant 'b' for the resulting gas;
(A) Increases (B) Decreases
(C) remains same (D) changes unpredictably

34. Given wave function represents which orbital of hydrogen $\psi = \frac{1}{4}\frac{1}{\sqrt{2\pi}}\left(\frac{1}{a_o}\right)^{5/2} re^{-r/2a_o\cos\theta}$ (where θ = angle from z-axis)
(A) $2P_y$ (B) $2P_z$
(C) $3P_y$ (D) $3P_z$

Space for Rough work

35. Two solid compounds X and Y dissociate at a certain temperature in a closed container as followed:

$X(s) \rightleftharpoons A(g) + 2B(g); k_{p_1} = 9 \times 10^{-3} atm^3$

$Y(s) \rightleftharpoons 2B(g) + C(g); k_{p_2} = 4.5 \times 10^{-3} atm^3$

The total pressure of gases over a mixture of X and Y at equilibrium is

(A) 4.5 atm (B) 0.45 atm

(C) 0.6 atm (D) None of these

36. Calculate resonance enthalpy of $CO_2(g)$ from the following data:

$\Delta H_C^o C(\text{graphite}) = -390 \text{ kJ/mol}$

$\Delta H_S^o C(\text{graphite}) = 715 \text{ kJ/mol}$

$\Delta H_{BE}^o [O = O] = 500 \text{ kJ/mol}$

$\Delta H_{BE}^o [C = O] = 875 \text{ kJ/mol}$

(A) -40 kJ/mol (B) -145 kJ/mol

(C) -72.5 kJ/mol (D) -290 kJ/mol

37. A sample containing $1 \text{ mol } KHC_2O_4.H_2C_2O_4$ is titrated with different reagent. Select correct statement.

(A) $\frac{1}{3}$ mol of KOH are used

(B) $3/2$ mol of $Ba(OH)_2$ are used

(D) $\frac{4}{5}$ mol of $KMnO_4$ are used in alkaline medium

(D) $\frac{1}{3}$ mol of $K_2Cr_2O_7$ are used in acidic medium

Space for Rough work

38. 1 mole of NH_3 gas at 27^oC is expanded under reversible adiabatic conditions to make volume 8 times $(\gamma = 1.33)$. Final temperature and work done respectively are

(A) 150 K, 900 cal
(B) 150 K, 400 cal
(C) 250 K, 1000 cal
(D) 200 K, 800 cal

39. In a compound, oxide ions are arranged in cubic close packing arrangement. Cations A occupy 1/6 of the tetrahedral voids and cations B occupy 1/3 of the octahedral voids. The formula of the compound is

(A) A_2BO_3
(B) AB_2O_3
(C) $A_2B_2O_2$
(D) ABO_3

40. Two substances $A(t_{1/2} = 5\text{min})$ and $B(t_{1/2} = 15\text{min})$ are taken in such a way that initially $[A] = 4[B]$. The time after which both the concentration will be equal is : (Assume that reaction is first order)

(A) 15 min
(B) 5 min
(C) 20 min
(D) concentration can never be equal

41. The reaction takes place in three steps with activation energy $E_{a_1} = 180\text{kJ/mol}$, $E_{a_2} = 80\text{kJ/mol}$, $E_{a_3} = 50\text{kJ/mol}$ respectively. Over all rate constant of the reaction is $k = \left[\frac{k_1 k_2}{k_3}\right]^{2/3}$ (k_1, k_2 and k_3 are rate constant of first, second and third step) . The overall activation energy will be

(A) 140 kJ/mol
(B) 150 kJ/mol
(C) 43.44 kJ/mol
(D) 100 kJ/mol

42. Bombardment of aluminum by $\alpha-$particle leads to its artificial disintegration in two ways (i) and (ii) as shown. Products x, y and z respectively are

$$\begin{array}{ccc} {}^{27}_{13}Al & \xrightarrow{(ii)} & {}^{30}_{15}P + Y \\ \downarrow (i) & & \downarrow \\ {}^{30}_{14}Si + X & & {}^{30}_{14}Si + Z \end{array}$$

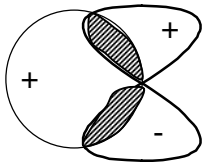
(A) proton, neutron, positron
(B) neutron, positron, proton
(C) proton, positron, neutron
(D) positron, proton, neutron

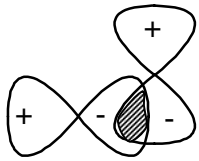
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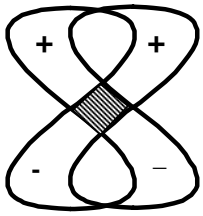
43. One mole of a compound $Co(NH_3)_5 Cl_2Br$ gives 2 moles of curdy white ppt. when treated with excess of $AgNO_3$ solution. Which of the following is incorrect about the compound?
(A) The compound may be represented as $\left[Co(NH_3)_5 Br\right]Cl_2$
(B) The Vant Hoff factor (i) is 3, when $\alpha = 1$
(C) The B.P of 1 M aq. Solution of $Co(NH_3)_5 Cl_2 Br$ should be triple of its value, when it were non electrolyte.
(D) The osmotic pressure of 1 M aq. Solution of $\left[CO(NH_3)_5 Br\right]Cl_2$ should be triple of its value when it were electrolyte.

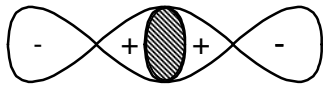
44. At 40^oC, vapour pressure in torr of methanol and ethanol solution is $P = 119X + 135$: where X is the mole fraction of methanol. Hence
(A) Vapour pressure of pure methanol is 119 torr.
(B) Vapour pressure of pure ethanol is 135 torr
(C) Vapour pressure of equimolar mixture of each is 127 torr.
(D) Mixture is completely immiscible.

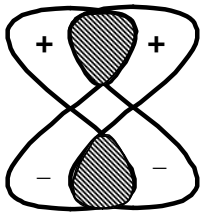
45. Which of the following atomic orbital overlapping are not allowed?

(P) 

(Q) 

(R) 

(S) 

(T) 

(A) All of these
(B) (P) (Q) (R)
(C) (P) (R) (T)
(D) (Q) only

Space for Rough work

46. The order of $p\pi - d\pi$ interaction in the compounds containing bond between Si/P/S/Cl and oxygen is in the order:
(A) $P > Si > Cl > S$ (B) $SI < P < S < Cl$
(C) $S < Cl < P < Si$ (D) $Si > P > S > Cl$

47. Select the **incorrect** statement.
(A) Br_2 and ICl have the same number of electrons and thus, both have nearly the same boiling points.
(B) N_2H_4 is pyramidal about each N-atom.
(C) In P_4S_3 molecule, there are six P–S bonds, three P–P bonds are then lone pairs of electrons (on all atoms).
(D) In ClO_4^-, all $Cl-O$ bonds are identical and there is strong $p\pi - d\pi$ bonding between chlorine and oxygen atoms.

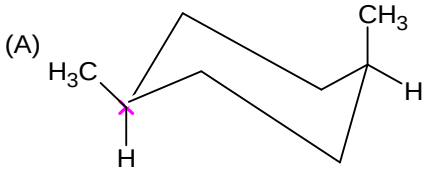
48. The diamagnetic complex which has two geometrical and 4 optically active isomers is
(A) $\left[Co(gly)_3\right]$ (B) $\left[Pt(en)_2Cl_2\right]^{2+}$
(C) $\left[Co(en)_3\right]^{3+}$ (D) $\left[Fe(Ox)_3\right]^{3-}$

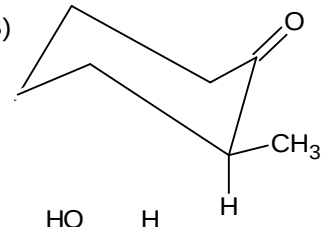
49. Among the following metal carbonyl carbonyls, the C–O bond is strongest in
(A) $\left[Mn(CO)_6\right]^+$ (B) $\left[Cr(CO)_6\right]$
(C) $\left[V(CO)_6\right]^-$ (D) $\left[Ti(CO)_6\right]^{2-}$

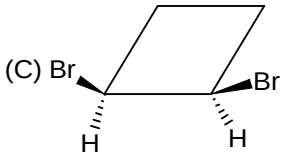
50. Which **does not** represent **correct** method?
(A) $TiCl_2 + 2Mg \rightarrow Ti + 2MgCl_2$: Kroll (B) $Ni(CO)_4 \rightarrow Ni + 4CO$ Mond
(C) $Ag_2CO_3 \rightarrow 2Ag + CO_2 + O_2$: Van Arkel (D) $ZrI_4 \rightarrow Zr + 2I_2$: Van Arkel

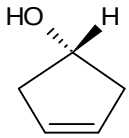
Space for Rough work

51. Which of the following molecules is chiral?

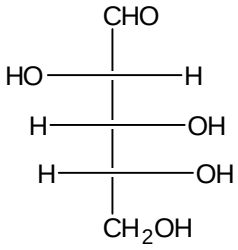
(A) 

(B) 

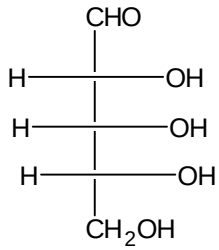
(C) 

(D) 

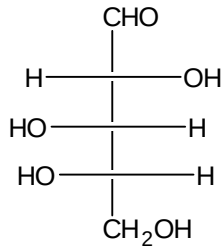
52. The correct statement about the compounds I, II and III:



I



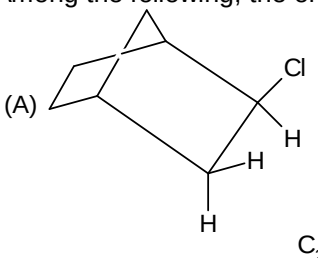
II



III

(A) II and III are epimers
(B) I and II are epimers
(C) I and III are diastereomers
(D) I and III are identical

53. Among the following, the one that undergoes syn-E2 reaction almost exclusively is

(A) 

$\xrightarrow[C_2H_5OH]{C_2H_5ONa}$

(B) $CH_3CH_2CH(Br)CH_3 \xrightarrow[C_2H_5OH]{KOH}$

(C) $CH_3CH_2CH(Me_2S^+I^-)CH_3 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$

(D) $(CH_3)_3CBr \xrightarrow[C_2H_5OH]{C_2H_5ONa}$

Space for Rough work

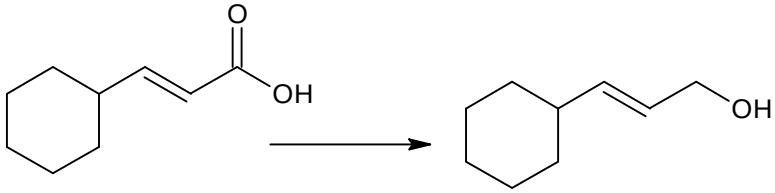
54. Consider the reaction

$CH_3CH=CH_2 \xrightarrow[2.\ CH_3COOH]{1.\ B_2D_6}$ The final product formed is

(A) CH_3CHDCH_3 (B) $CH_3CH_2CH_2D$

(C) CH_3CHDCH_2D (D) $CH_3CHDCH_2OCOCH_3$

55. In order to achieve the conversion

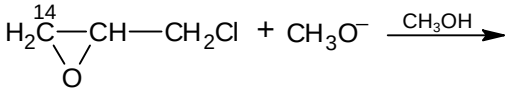


Which of following reducing agents should be employed?

(A) $NaBH_4$ (B) H_2, Ni

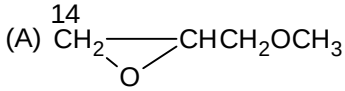
(C) $LiAlH_4$ (D) $\left[(CH_3)_2CHO\right]_3Al,(CH_3)_2CHOH$

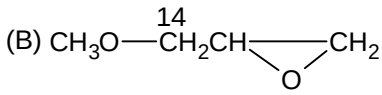
56. For the reaction:

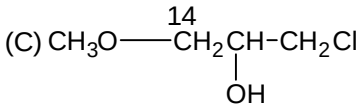


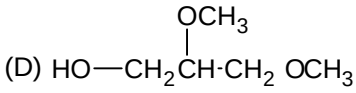
(A) major product.

(A) is

(A) 

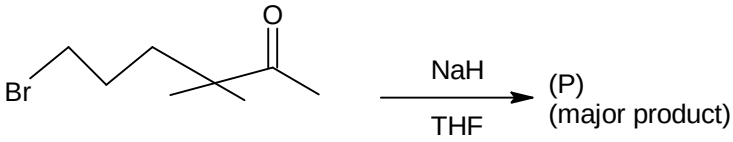
(B) 

(C) 

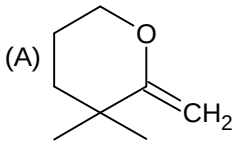
(D) 

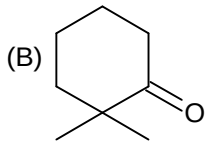
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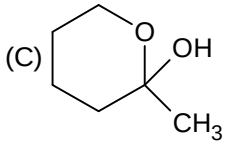
57. For the reaction

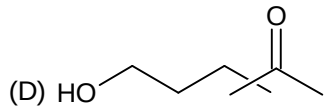


the major product (P) is

(A) 

(B) 

(C) 

(D) 

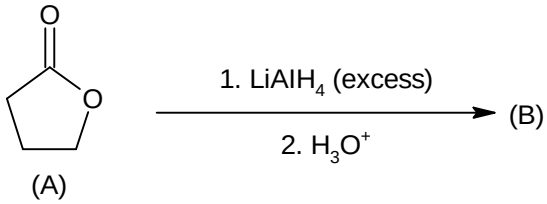
58. Which of the following carboxylic acids undergoes decarboxylation most readily on heating?

(A) $C_6H_5CH(NH_2)COOH$ (B) $C_6H_5CH(OH)COOH$

(C) $C_6H_5COCOOH$ (D) $C_6H_5COCH_2COOH$

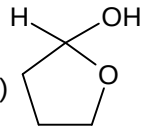
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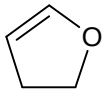
59. γ – Butyrolactone, (A), is reduced with $LiAlH_4$;



(A)

The product (B) is

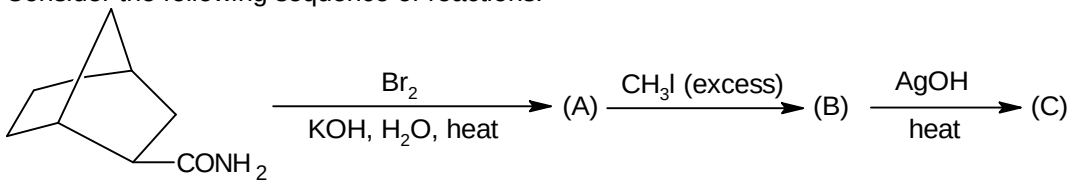
(A) 

(B) 

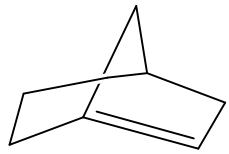
(C) $HOCH_2CH_2CH_2OH$

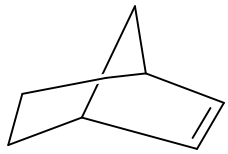
(D) $CH_3CH_2CH_2CH_2OH$

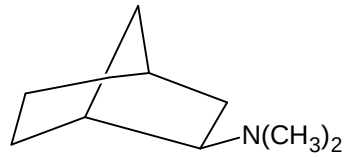
60. Consider the following sequence of reactions.

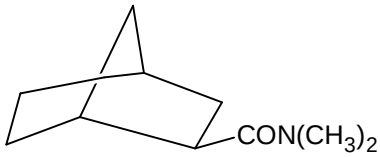


The end product (C) is

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If equation of hyperbola touching x-axis and lines $x-2y=0$, $x+y+1=0$ as its asymptotes is $\alpha(x-2y)(x+y+1)+\beta=0$ $(\alpha,\beta\in N$ and their H.C.F is $1)$, then value of $\alpha-\beta$ is

(A) 1 (B) 2
(C) 3 (D) 4

62. Let $f(y)=\int_0^1 e^{x-y^2}\left(x^3-2xy+3x^2-2y\right)dx$, then $\int_0^1 e^y f(y)dy$ is equal to

(A) 1 (B) $e-1$
(C) 0 (D) e^2-2e

63. If $x^2-2x+2\sin\alpha=0$ has unique root in $(-1,1)$, then length of largest continuous interval of α in $[0,2\pi]$ is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
(C) $\frac{5\pi}{6}$ (D) $\frac{7\pi}{6}$

64. Difference of maximum and minimum value of $f(x)=\sqrt{\cos^{-1}[x-2]}$ is $([.]$ is G.I.F$)$

(A) $\sqrt{\pi}$ (B) $\frac{\sqrt{\pi}}{2}$
(C) $\sqrt{\pi}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$ (D) π

Space for Rough work

65. Let K is number of outcomes by throwing 5 identical dice then maximum values of λ for which 2^{λ} divides K is

(A) 2 (B) 3
(C) 4 (D) 6

66. Tangent to parabola $y^2 = 4p(x+1)$ cuts hyperbola $xy = 2$ in A and B, then locus of mid points of AB is

(A) $\frac{x}{y} + (2x-1) = 0$ (B) $\frac{x^2}{y^2} + 2x - 1 = 0$

(C) $\frac{x}{y} + (2x-1)^2 = 0$ (D) $\frac{x^2}{y^2} + (2x+1) = 0$

67. Let $|z+i| - z + i = \theta$, then arrangements of z can't be

(A) $-\frac{\pi}{2}$ (B) $-\frac{\pi}{4}$

(C) $-\frac{\pi}{6}$ (D) $-\frac{3\pi}{4}$

68. There are two bags, bag 1 contains 3 white and 2 red balls and bag 2 contains 4 white and 5 red balls. A ball is drawn randomly from bag 1 and put in bag 2. Now 2 balls are drawn from bag 2 and found to be red. Then the probability that white ball was drawn from bag 1 to bag 2 is

(A) $< \frac{1}{2}$ (B) $< \frac{1}{4}$

(C) $> \frac{3}{4}$ (D) $> \frac{1}{4}$

69. Value of $\frac{5}{(2+4)^2} + \frac{7}{(2+4+6)^2} + \frac{9}{(2+4+6+8)^2} + ...\infty$ is

(A) $\frac{1}{9}$ (B) $\frac{1}{8}$

(C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Space for Rough work

70. Let $f : R \to R$ such that $f(x + 2y) = f(x) + f(2y) + 4xy$, $x, y \in R$ and $f'(0) = 0$.
If $I_1 = \int_0^1 f(x)dx, I_2 = \int_{-1}^0 f(x)dx, I_3 = \int_{1/2}^2 f(x)dx$, then
(A) $I_1 = I_2 > I_3$ (B) $I_1 > I_2 > I_3$
(C) $I_1 = I_2 < I_3$ (D) $I_1 < I_2 < I_3$

71. Value of $\lim\limits_{n\to\infty} \sum_{1}^{4n} \frac{\sqrt{n}}{\sqrt{r}\left(3\sqrt{r} + 4\sqrt{n}\right)^2}$ is equal to
(A) $\frac{1}{35}$ (B) $\frac{1}{14}$
(C) $\frac{1}{10}$ (D) $\frac{1}{5}$

72. Normal to a curve at a point P (x, y) meets the x-axis at G. If distance of G from the origin is twice the abscissa of P, then the curve is
(A) Parabola (B) Circle
(C) Ellipse (D) Hyperbola

73. The area inside parabola $5x^2 - y = 0$ but outside the parabola $2x^2 - y + 9 = 0$ is
(A) $12\sqrt{3}$ sq. unit (B) $6\sqrt{3}$ sq. unit
(C) $8\sqrt{3}$ sq. unit (D) $4\sqrt{3}$ sq. unit

74. Minimum value of $n\sin\alpha + \frac{1}{n}\cos\alpha$, where $n > 0$ and α is fixed in (0, 1), is
(A) $\sqrt{n^2 + n^{-2}}$ (B) $\sqrt{\sin 2\alpha}$
(C) 0 (D) $\sqrt{2\sin 2\alpha}$

75. Let A and B are two events. If $P(A) = 0.4, P(B / A) = 0.9, P(\overline{B} / \overline{A}) = 0.6$, then which is not correct?
(A) $P(A \cup B) = .64$ (B) $P(A \cap B) = .36$
(C) $P(A / B) = 0.6$ (D) $P(A / \overline{B}) = 0.2$

Space for Rough work

76. If $y = mx$ bisect two chords of $y^2 = 4x$ from $(4, 4)$, then m can't be

(A) $m = \frac{3}{4}$ (B) $m = \frac{1}{2}$

(C) $m = \frac{5}{6}$ (D) $m = -\frac{1}{2}$

77. Number of circular arrangement of 11 identical green balls, 1 red, 1 white and 1 blue ball is

(A) ${}^{13}P_2$ (B) ${}^{13}P_{11}$

(C) ${}^{14}C_2$ (D) ${}^{14}P_{11}$

78. If $f(x)$ be differentiable function and curve $y = f(x)$ passes through (1, 1) and satisfies the relation $2f(x+y) + f(x-y) + 3y^2 = 3f(x) + 2xy$, then $\lim_{x\to 1}\frac{f(x)-1}{x-1}$ is equal to

(A) 3 (B) 0

(C) 2 (D) 1

79. Given two real sets $A = \{a, a_2, a_3 \cdots a_{2n}\}$ and $B\{b_1, b_2, \cdots b_n\}$. If $f : A \to B$ is a function such that every element of B has an inverse image and $f(a_1) \le f(a_2) \le f(a_3) \le f(a_4) \cdots \le f(a_{2n})$, then the number of such mappings are

(A) ${}^{2n}C_n$ (B) ${}^{2n}C_{n-1}$

(C) ${}^{2n-1}C_{n-1}$ (D) ${}^{2n+1}C_n$

80. If A = 30° and area of triangle ABC is $\frac{\sqrt{3}}{4}a^2$, then the triangle ABC is

(A) Right angled (B) Acute angled

(C) Obtuse angled (D) None of these

81. If α, β are roots of equation $ax^2 + bx + c = 0$, then the system of equation

$x + y\cos(\beta - \alpha) + z\cos\alpha = 0$

$x\cos(\alpha - \beta) + y + z\cos\beta = 0$

and $x\cos\alpha + y\cos\beta + z = 0$ has

(A) only trivial solution (B) has non trivial solution

(C) unique solution (D) exactly 3 solution

Space for Rough work

82. If A,B,C are angles of a triangle such that $\lim\limits_{x\to\infty}\left[\frac{\sin\left(Ax^2+Bx+C\right)}{\sin\left(Cx^2+Bx+A\right)}+\frac{\cos\left(Ax^2+Cx+B\right)}{\cos\left(Cx^2+Ax+B\right)}\right]=\frac{2}{\sqrt{3}}$, $\angle B=30^\circ$ and perimeter of triangle is $4+2\sqrt{3}$ then area of triangle is

(A) $\sqrt{3}$ sq. unit (B) 2 sq. unit

(C) $\frac{1}{2}$ sq. unit (D) $3\sqrt{3}$ sq. unit

83. Consider the polynomial $f(x)=2x^4-ax^3+2bx^2+cx-d$, where $a,b,c,d\in R$ and $f(i)=f(3+i)=0$, then the value of b + d is

(A) -5 (B) -9

(C) 11 (D) 7

84. If z is a complex number such that $\arg z(1+\bar{z})+\arg\left(\frac{|z|^2}{z-|z^2|}\right)=0$, then

(A) $\arg\bar{z}=-\frac{\pi}{2}$ (B) $\arg z=\frac{\pi}{4}$

(C) $|\bar{z}|<1$ (D) $\ln\left(\frac{1}{|z|}\right)\in(-\infty,\infty)$

85. If $\sum\limits_{n=1}^{k}\left[\frac{1}{3}+\frac{n}{90}\right]=21$, where [x] denotes the integral part of x, then k is equal to

(A) 84 (B) 80

(C) 85 (D) 86

86. A hat contains a number of cards with 30% white on both sides, 50% black on one side and white on the other side, 20% black on both sides. The cards are mixed up, and a single card is drawn at random and placed on the table. Its upper side shows up black. The probability that its other side is also black is

(A) 2/9 (B) 4/9

(C) 2/3 (D) 2/7

Space for Rough work

87. A man is moving away from a tower 41.6 m high at a rate of 2 m/s. If the eye level of the man is 1.6 m above the ground, then the rate at which the angle of elevation of the top of the tower changes, when he is at a distance of 30 m from the foot of the tower, is

(A) $-\frac{4}{125}$ radian/s (B) $-\frac{2}{25}$ radian/s

(C) $-\frac{1}{625}$ radian/s (D) None of these

88. The value of the expression $2(1+\omega)(1+\omega^2)+3(2\omega+1)(2\omega^2+1)+4(3\omega+1)(3\omega^2+1)+.....$ $.....+(n+1)(n\omega+1)(n\omega^2+1)$ is (ω is a complex cube root of unity)

(A) $\frac{n^2(n+1)^2}{4}$ (B) $\left(\frac{n(n+1)}{2}\right)^2+n$

(C) $\left(\frac{n(n+1)}{2}\right)^2-n$ (D) $\frac{n^2(n+1)^2}{2}$

89. The number of solutions of $\sec^2\theta\,\text{cosec}^2\theta+2\,\text{cosec}^2\theta=8,\ 0\le\theta\le\pi/2$ is

(A) 4 (B) 3

(C) 0 (D) 2

90. $\sum_{m=1}^{n}\tan^{-1}\left(\frac{2m}{m^4+m^2+2}\right)$ is equal to

(A) $\tan^{-1}\left(\frac{n^2+n}{n^2+n+2}\right)$ (B) $\tan^{-1}\left(\frac{n^2-n}{n^2-n+2}\right)$

(C) $\tan^{-1}\left(\frac{n^2+n+2}{n^2+n}\right)$ (D) None of these

Space for Rough work